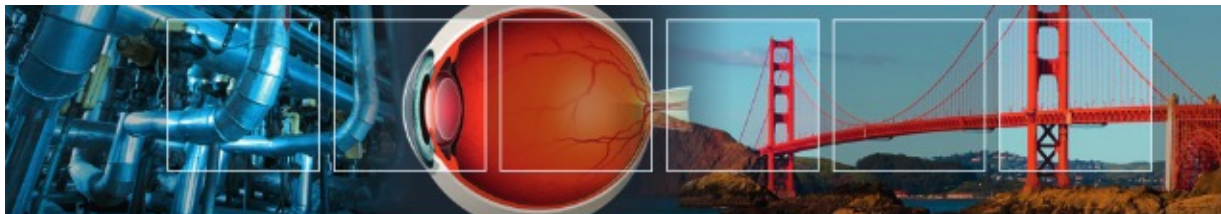


Localisation



Overview

- Without knowledge of location of a sensor, the information produced by such sensor is of limited use
 - location of sensed events in the physical world
 - location-aware services
 - location is often the primary sensor information (object tracking, supply chain management, surveillance)
- Location is often not known knowledge, therefore, localization is the task of determining the position (e.g., coordinates) of a sensor or the spatial relationships among objects
- Application needs determine technology
 - Accurate
 - Low-cost
 - Easy-to-deploy
 - Ubiquitous



Typical positioning techniques

- Network based – main challenge is multi-path, environment fluctuation
 - WiFi, Non-WLAN including BLE, 802.15.4 (Zigbee, 6LowPAN), UWB, RFID, NFC
 - Radio wave propagation between transmitter and receiver
 - RSSI, ToA, TDoA, AoA/DoA, CIR, CSS
 - Algorithms involved in position estimation
 - Trilateration, Triangulation, Fingerprinting, Proximity (kNN, Machine Learning)

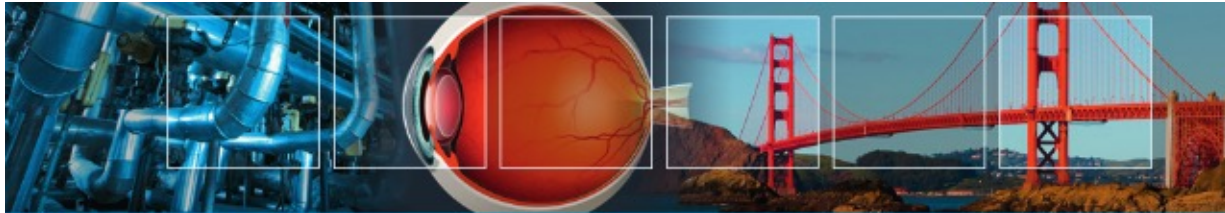


Typical positioning techniques

- Sensor based – main challenge is sensor drift
 - Accelerometer, Gyroscope, Magnetometer, Barometer, Camera (stereo vision), distance-based sensors (Ultrasonic, Laser, Odometer)
 - Sensor data characterizing movement, heading, height
 - Acceleration, Angular acceleration/orientation, Magnetic field, air pressure, stereo vision, distance
 - Algorithms involved in position estimation
 - Double integration, dead reckoning, image recognition
- Applicable to both human and non-human objects
 - Pedestrian tracking – walking pattern may vary significantly
 - Locating goods in warehouse, robots



Wireless Localisation



Wireless Technologies for Localization

Name	Effective Range	Pros	Cons
GSM	35km	Long range	Very low accuracy
LTE	30km-100km		
Wi-Fi	50m-100m	Readily available; Medium range	Low accuracy
Ultra Wideband	70m	High accuracy	High cost
Bluetooth	10m	Readily Available; Medium accuracy	Short range
Ultrasound	6-9m	High accuracy	High cost, not scalable
RFID & IR	1m	Moderate to high accuracy	Short range, Line-Of-Sight
NFC	<4cm	High accuracy	Very short range



Wireless Localization Techniques

■ Range-based algorithms

- Require determining the range (i.e. distance)

■ Fingerprinting

- Require site surveying
- May require other statistical/machine learning algorithms' assistance
- May be zone based (i.e. low location resolution)



Ranging Techniques

- Time of Arrival (ToA, time of flight)
 - distance between sender and receiver of a signal can be determined using the measured signal propagation time and known signal velocity
 - sound waves: 343m/s, i.e., approx. 30ms to travel 10m
 - radio signals: 300km/s, i.e., approx. 30ns to travel 10m



Ranging Techniques

■ One-way ToA

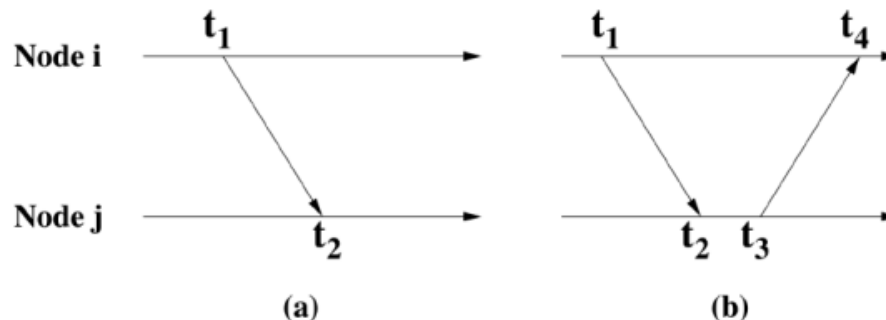
- one-way propagation of signal
- requires highly accurate synchronization of sender and receiver clocks

$$dist_{ij} = (t_2 - t_1) * v$$

■ Two-way ToA

- round-trip time of signal is measured at sender device
- third message if receiver wants to know the distance

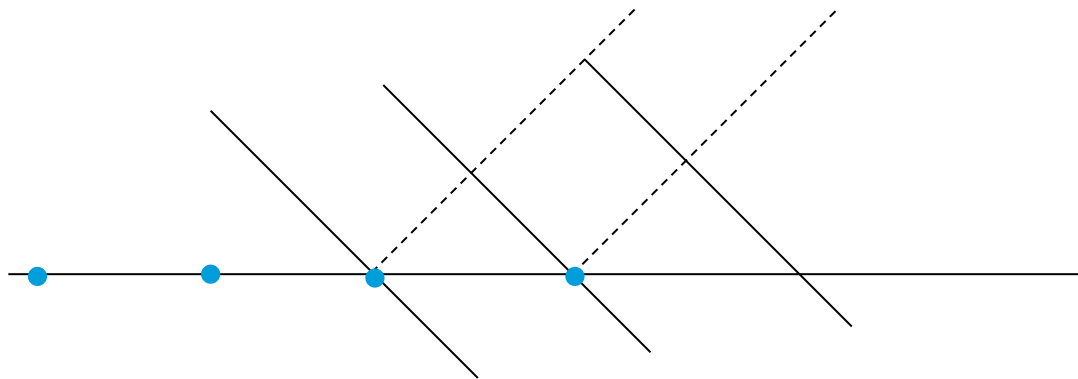
$$dist_{ij} = \frac{(t_4 - t_1) - (t_3 - t_2)}{2} * v$$



Ranging Techniques

■ Angle of Arrival (AoA)

- direction of signal propagation
- typically achieved using an array of antennas or microphones
- angle between signal and some reference is orientation
- spatial separation of antennas or microphones leads to differences in arrival times, amplitudes, and phases
- accuracy can be high (within a few degrees)
- adds significant hardware cost



Ranging Techniques

■ Received Signal Strength (RSS)

- many devices measure signal strength with received signal strength indicator (RSSI)
 - ▶ vendor-specific interpretation and representation
 - ▶ typical RSSI values are in range of 0..RSSI_Max
- in free space, RSS degrades with square of distance, expressed by Friis transmission equation

$$\frac{P_r}{P_t} = G_t G_r \frac{\lambda^2}{(4\pi)^2 R^2}$$

- One simplest form of converting RSSI to distance is:

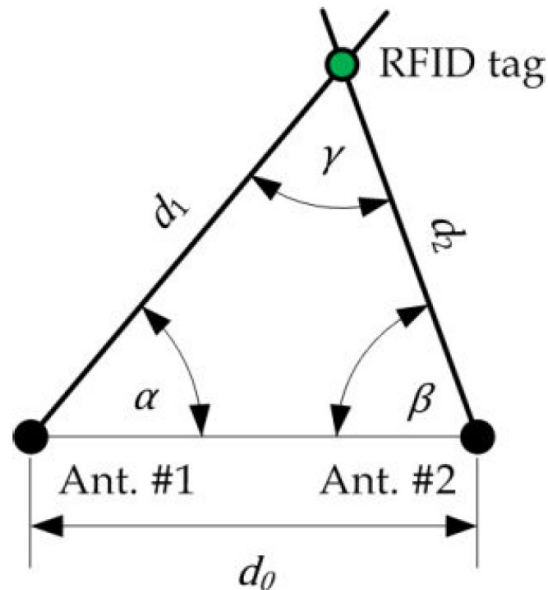
$$\text{RSSI (dBm)} = -10n \log_{10}(d) + A$$

- ▶ A is the RSSI in dBm at 1 meter (a reference distance)
- ▶ n is a path loss exponent (or propagation constant). For free space, n=2
- in practice, the actual attenuation depends on multipath propagation effects, reflections, noise, etc.



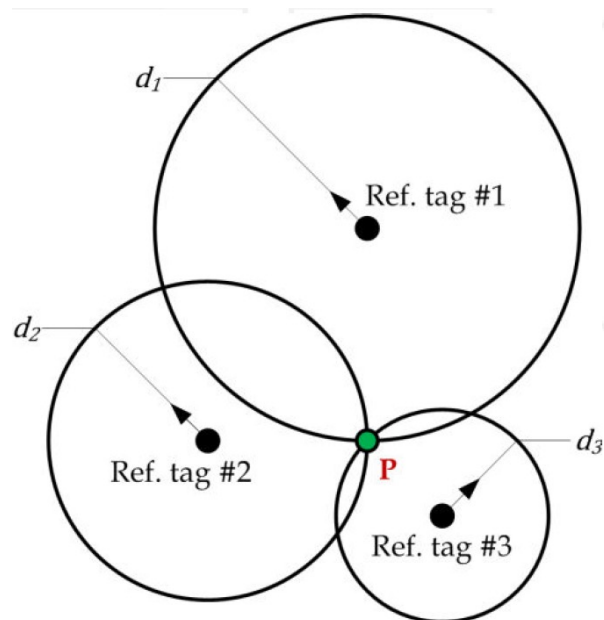
Triangulation

- Example of range-based localization
- Uses the geometric properties of triangles to estimate location
- Relies on angle (bearing) measurements
- Minimum of two bearing lines (and the locations of anchor nodes or the distance between them) are needed for two-dimensional space



Trilateration

- Localization based on measured distances between a node and a number of anchor points with known locations
- Basic concept: given the distance to an anchor, it is known that the node must be along the circumference of a circle centred at anchor and a radius equal to the node-anchor distance
- In two-dimensional space, at least three non-collinear anchors are needed and in three-dimensional space, at least four non-coplanar anchors are needed



Fingerprinting

- Mapping solution
- Address problems with multipath
- Better than modeling complex RF propagation pattern
- Step1
 - Build up location fingerprints (i.e. location coordinates + respective RSSI from nearby base stations)
- Step2
 - Match online measurements with the closest a priori location fingerprints
- Weakness
 - Need prior site survey for every site
 - Any change to the physical environment requires the survey to be conducted again

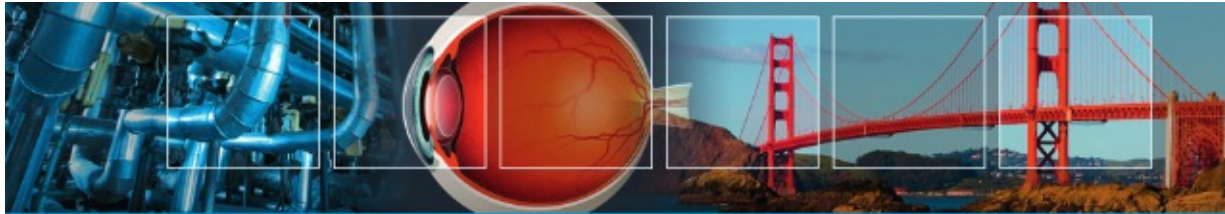


Summary of Localization Techniques

	Measurement Scheme	Accuracy	Special Requirement
Range-based	TOA	Moderate	Synchronization, dense beacons
	TDOA	High	Synchronization, LOS, dense beacons
	AOA	High	Directional antenna
	RSSI	Moderate	No
Fingerprinting	RSSI	High	Site survey



Inertial Sensor (INS) Localisation



Applications of INS

- Mechanical Inertial Navigation Systems (INS) are already applied in many application domains
 - Automotive
 - Aeronautics and Space: Auto-pilot, helicopter, space shuttle
 - Robotics: Unmanned navigation
- Based on measurements from accelerometer and gyroscope to provide position and orientation information
- INS are
 - Self-contained
 - Traditionally bulky but recent MEMS technology makes INS become more popular
 - MEMS IMU (Inertial Measurement Unit) has low accuracy



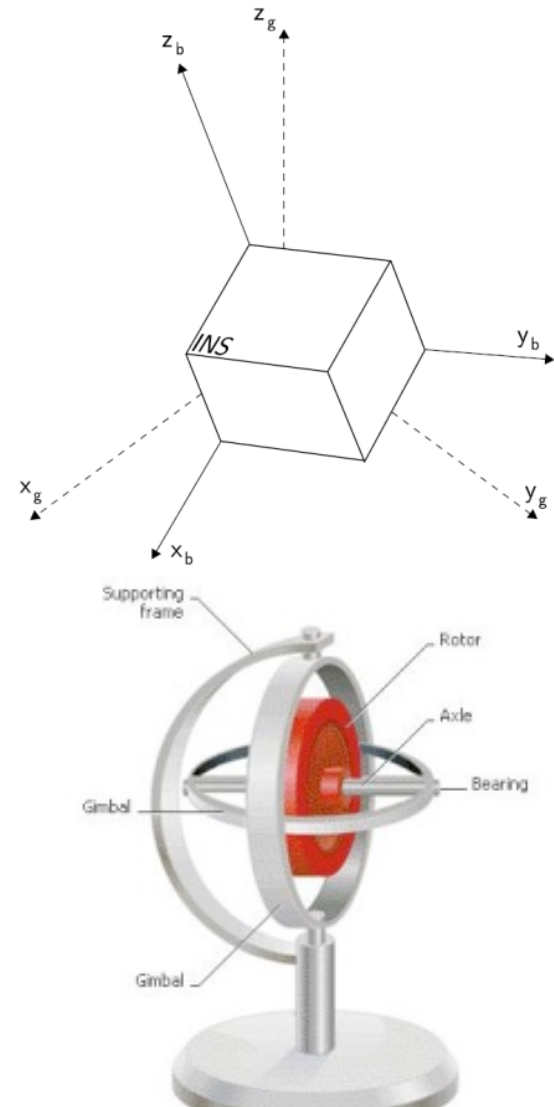
Type of INS

■ Stable systems

- Sensors are mounted on a platform where external motion is isolated
- Aligned with global frame via mounting on gimbal frames

■ Strapdown (our focus on indoor positioning)

- Sensors are mounted on INS devices
- Measurements are based on body frame rather than global frame



Accelerometer

■ Newton's Second Law

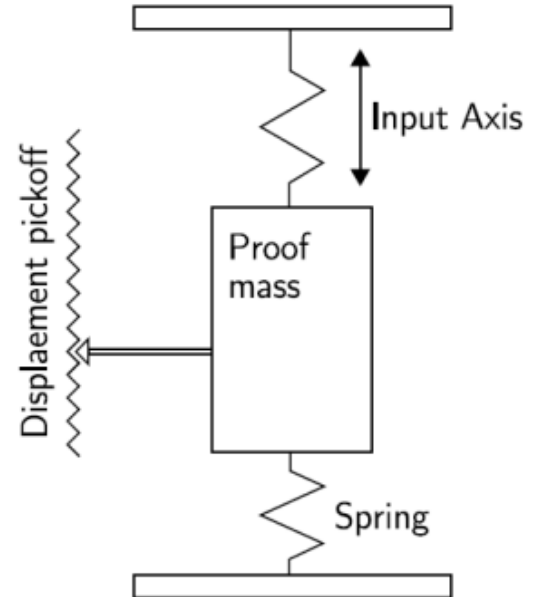
- $F = mA$

■ Measure force on object of known mass (proof mass) to determine acceleration

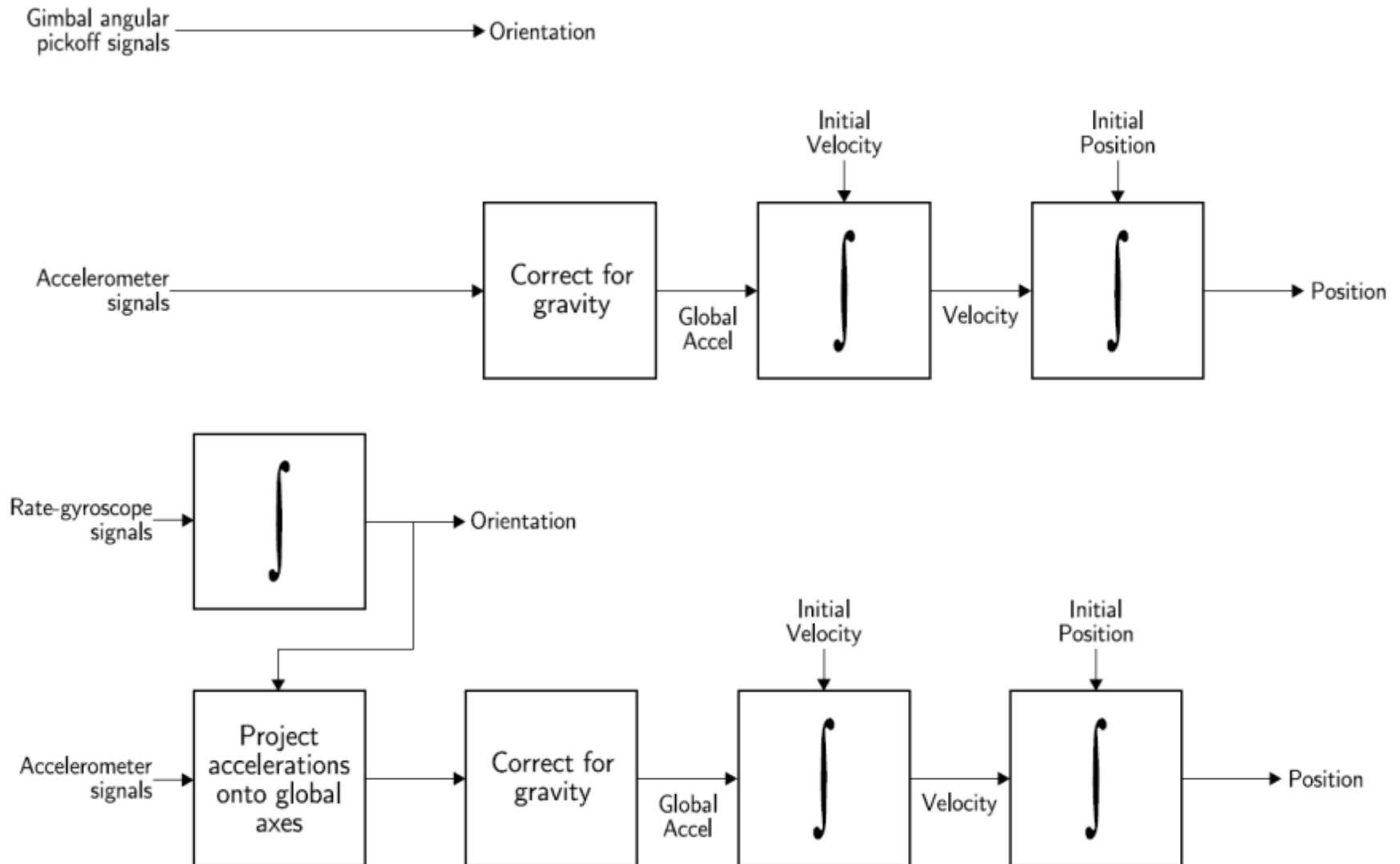
- Measuring displacement
- Measuring change in vibration frequency caused by change in tension force

■ Need to consider gravity

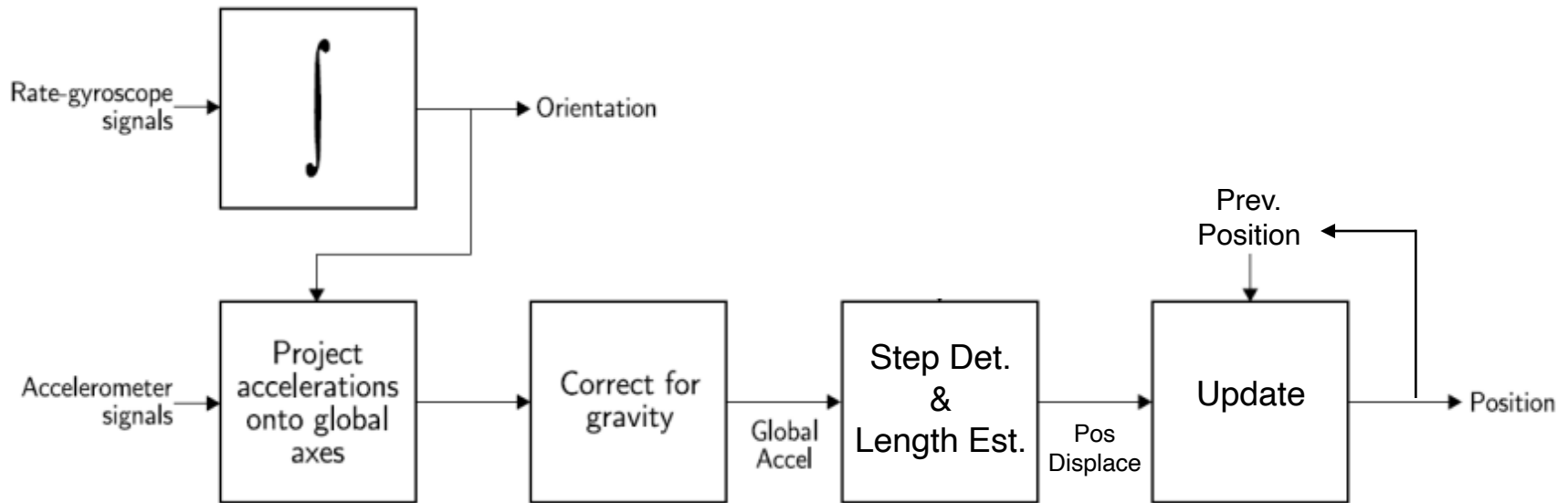
- $A_{\text{measured}} = A_{\text{actual}} + g$
- Removing gravity components for 3D accelerometer can be more complex



Inertial Navigation – Double Integration



Inertial Navigation – Step Detection



■ Step detection

- Peak detection, zero-crossing detection

■ Step length estimation

- h – height, a – vertical acceleration in global frame, f_s – step frequency

$$L_s = k_1 * h$$

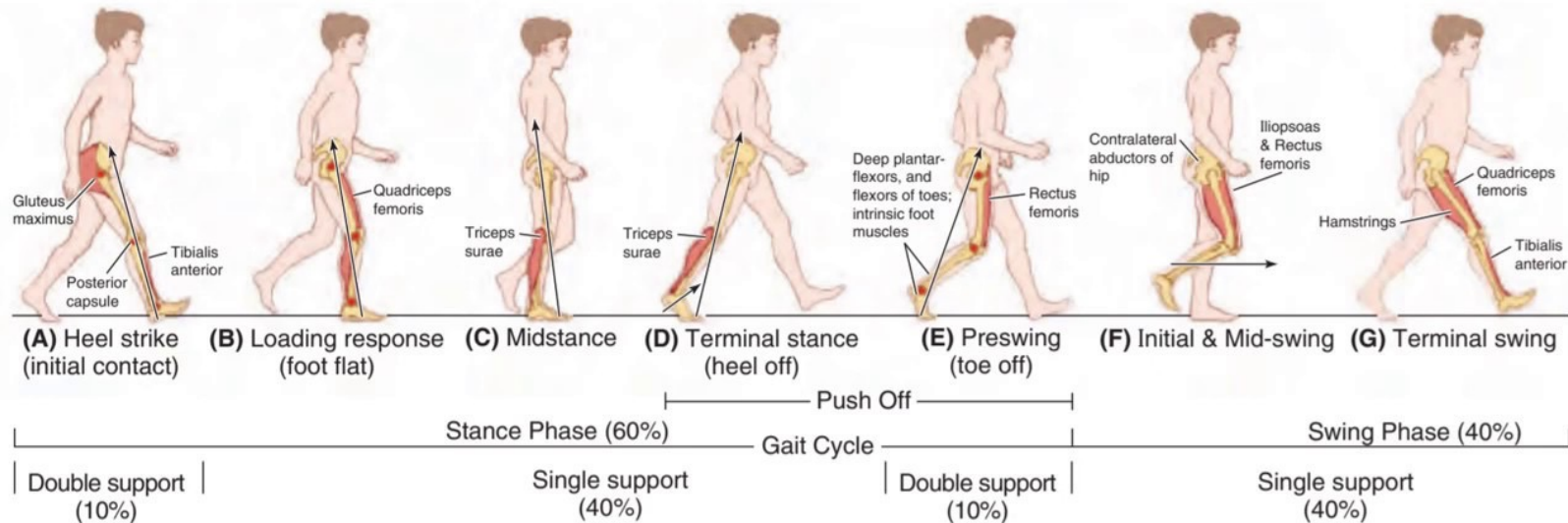
$$L_s = k_2 * \sqrt[4]{a_{max} - a_{min}}$$

$$L_s = k_3 * \sqrt[3]{\sum_{i=1}^N |a_i| / N}$$

$$L_s = k * h * \sqrt{f_s}$$



Inertial Navigation – Step Detection



■ Step detection

- Peak detection, zero-crossing detection

■ Step length estimation

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Moore KL, Daley II AF, Agur AMR. *Clinically Oriented Anatomy* [ebook]. 7th ed. Baltimore: Wolters Kluwer; 2014. Figure 5.20, 'Gait cycle'.

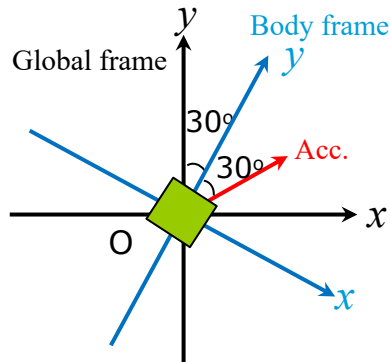
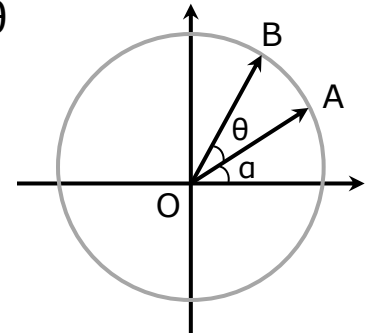


Inertial Navigation - Rotation

- Point A and B are on unit circle, the coordinates of A and B are $(\cos\alpha, \sin\alpha)$ and $(\cos(\alpha+\theta), \sin(\alpha+\theta))$, respectively

- $\cos(\alpha+\theta) = \cos\alpha\cos\theta - \sin\alpha\sin\theta$, $\sin(\alpha+\theta) = \cos\alpha\sin\theta + \sin\alpha\cos\theta$

- Written in matrix form
$$\begin{bmatrix} \cos(\alpha + \theta) \\ \sin(\alpha + \theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}$$



$$\begin{bmatrix} \cos(-30^\circ) & -\sin(-30^\circ) \\ \sin(-30^\circ) & \cos(30^\circ) \end{bmatrix} * \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} * \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$$

Acc. of magnitude 1, the angle between the direction and body frame y is 30° , the Acc. vector in body frame can be written as $(1/2, \sqrt{3}/2)$; the angle between the direction and global frame y is 60° , the coordinate of the vector in global frame is $(\sqrt{3}/2, 1/2)$.



Inertial Navigation - Orientation

- For a strapdown INS, assuming an object is moving in x-y plane (i.e. 2D)
 - Acceleration will be measured in two components along the body frame x-y axes
 - Orientation (heading direction) can be measured via the angular velocity along the axes perpendicular to the x-y plane
- Assuming the angle between body frame, b , and global frame p is ψ_{pb} , then the acceleration components can be mapped to global frame via multiplication with a transformational matrix (or rotation matrix)

$$\begin{pmatrix} a_{pb,x}^p(t') \\ a_{pb,y}^p(t') \end{pmatrix} = \begin{pmatrix} \cos \psi_{pb}(t') & -\sin \psi_{pb}(t') \\ \sin \psi_{pb}(t') & \cos \psi_{pb}(t') \end{pmatrix} \begin{pmatrix} a_{pb,x}^b(t') \\ a_{pb,y}^b(t') \end{pmatrix}$$

- Questions: Is 2D sufficient?
 - Can 2D handle tilting? Or can you keep the device perfectly levelled? Is this practical?
 - If there is tilting, what happens with the gravity component?



A step-based INS demo



Project considerations

- In the project, you are given a device with accelerometer and magnetometer
- You want to design a system such that the device can track/trace user movements in time
- What is the dynamic range of the used sensors
 - Is the dynamic range configurable?
 - Is the dynamic range suitable for my application?
- How to interpret the sensor readings?
- Do I need all sensor data?
- Is the sensor data good quality?
 - If not, what to do? (calibration, compensation?)
- How to detect a step and orientation?
- How to track the movement (displacement and orientation) in time?

